A catalog of archaeal HTH and RHH DNA-binding-domain families outside a nineteen-family reference set

Ajay Vellanki[1],* and Shaurya Dosapati[1],*

[1]Independent researchers, West Windsor, New Jersey, United States of America

*Correspondence: ajayvellanki0720@gmail.com; devshaurya25@gmail.com

Ajay Vellanki ORCID: 0009-0008-7994-152X

These authors contributed equally.

Keywords: Archaea; DNA-binding domain; helix-turn-helix; ribbon-helix-helix; comparative genomics; protein structure

Abstract

Archaeal regulatory annotation relies heavily on familiar helix-turn-helix and ribbon-helix-helix families, leaving the organization of related sequences unclear. We asked how family-level diversity within Pfam clans CL0123 and CL0057 was organized after removal of an historical nineteen-family reference set. A lineage-aware workflow applied to 331 genomes produced 250 candidate families containing 13,237 proteins. Random sequence holdout gave high profile-HMM assignment accuracy, whereas complete-genus holdout across the deposited sequence collection yielded a median family accuracy of 82.6% and left five families unevaluable. This attenuation shows that taxonomic separation is necessary when assessing family boundaries. Fixed family profiles also transferred to independent genome accessions. Upstream-sequence signals weakened when genera or major lineages were held-out, and broader database and structural searches left nine initially uncharacterized families unresolved. The original profile inventory, raw discovery hits and corrected 3,526-genome family matrices were not preserved, so the historical discovery stage is not completely reproducible. The resulting versioned catalog organizes a bounded part of archaeal DNA-binding-domain diversity and prioritizes families for experimental testing; it does not establish transcription-factor activity.

Introduction

Archaea combine a eukaryote-like basal transcription apparatus with many compact bacterial-type regulators. Most annotated archaeal one-component regulators are assigned to a familiar set of HTH, RHH, and metal-

responsive families [1–6]. That reference set is useful, but it can also make the recognizable fraction appear more complete than it is.

A protein that carries a small DNA-binding fold is not necessarily a transcription factor. Related domains occur in antitoxins, defense and mobile element proteins, chromosome proteins, and enzymes involved in DNA transactions. Sequence, structure, genomic-neighborhood, motif and phylogenetic evidence therefore answer different questions. In this study, DNA-binding domain protein denotes a sequence assigned to the targeted HTH or RHH Pfam clans. Candidate regulator denotes a family with additional contextual evidence. Transcription factor is reserved for experimentally demonstrated regulatory activity.

Our question is deliberately narrower than a survey of every archaeal DNA-binding protein. We examine members of Pfam clans CL0123 and CL0057 that were not assigned to an historical nineteen-family reference set, group them into families, map their distribution and prioritize families for experiments. Proteins outside these clans, highly diverged sequences missed by their profiles and non-HTH or non-RHH DNA-binding mechanisms are outside the search universe. The original mapping of nineteen families to twenty Pfam models and the 747-model candidate inventory were not preserved. Accordingly, outside the reference set is an operational historical label rather than a reproducible claim about present-day Pfam coverage.

Results

A rule-based discovery panel

The discovery panel was generated by an explicit rule rather than by selecting a target of 331 genomes. All genomes available for Thaumarchaeota within TACK, DPANN, Asgardarchaeota, other deep TACK lineages and Korarchaeota within TACK were retained. Euryarchaeota and Crenarchaeota were capped at 140 and 70 genomes, respectively; genomes within those two groups were sampled in proportion to subgroup abundance with random seed 42. The rule produced 331 genomes: 140 Euryarchaeota, 70 Crenarchaeota within TACK, 51 Thaumarchaeota within TACK, 33 DPANN, 19 Asgardarchaeota, 15 other deep TACK, and 3 Korarchaeota within TACK. The caps are pragmatic controls for database imbalance rather than biologically optimal values. Panel composition and the selection rule are summarized in Table 1.

Table 1. Discovery-panel composition

Major group	Genomes	Panel rule
Euryarchaeota	140	Proportional sampling within group; cap 140
Crenarchaeota within TACK	70	Proportional sampling within group; cap 70
Thaumarchaeota within TACK	51	All available genomes retained
DPANN	33	All available genomes retained
Asgardarchaeota	19	All available genomes retained
Other deep TACK	15	All available genomes retained
Korarchaeota within TACK	3	All available genomes retained

The panel includes 155 complete assemblies, 4 chromosome-level assemblies, 58 scaffolds, and 114 contig-level assemblies. This mixture broadens lineage representation but makes absence calls conservative, especially in incomplete genomes.

A bounded search reveals diversity outside the historical reference set

We scanned the 331 proteomes against profiles from Pfam clans CL0123 and CL0057 after removing twenty profile models reported to represent nineteen reference families. Candidate and reference matches were compared on the aligned protein-domain interval. For each protein, the highest-scoring candidate interval was retained. A reference hit excluded that candidate only when it overlapped more than half of the candidate interval and its domain bit score was no more than 5 bits below the candidate score. The 5-bit tolerance was inherited rather than prospectively optimized. The exact historical Pfam release, reference profile accessions, candidate profile accessions and complete prefilter hit tables are unavailable. This stage therefore defines the provenance boundary of the release and cannot be rerun exactly.

Within this defined universe, 38,153 proteins carried a targeted domain. Of these, 16,535 (43.3 percent) matched the nineteen-family reference set, 14,451 (37.9 percent) matched another characterized family, and 7,167 (18.8 percent) were assigned to a DUF or lacked a Pfam annotation. These numbers quantify diversity relative to the chosen reference and clans; they do not imply that 56.7 percent of all archaeal transcription factors are unknown.

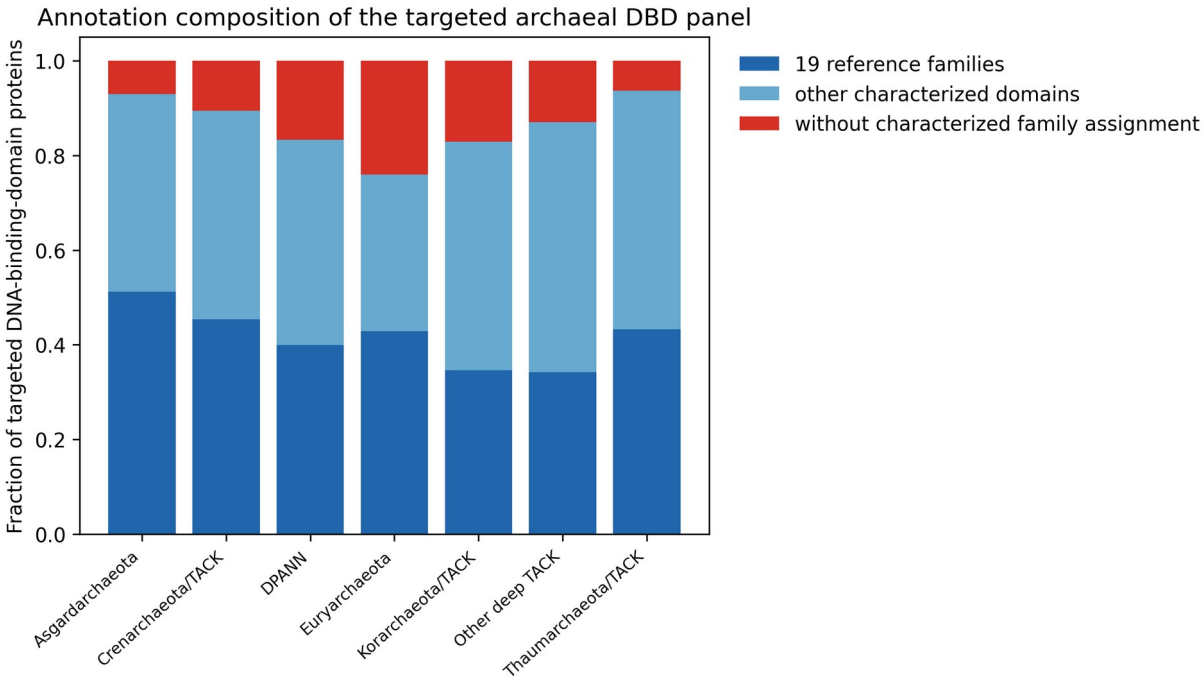


Figure 1. Annotation composition of the targeted archaeal DBD panel. Fractions are calculated only among proteins detected through Pfam clans CL0123 and CL0057. “Without characterized family assignment” is an operational annotation category, not evidence of a new fold or function.

Sequence clustering defines 250 candidate families

The noncanonical proteins were clustered with MMseqs2 at 30 percent sequence identity and 60 percent coverage. We retained clusters with a median protein length below 250 amino acids, at least 60 percent of members below 250 amino acids, and occurrence in at least five panel genomes, then removed clusters

carrying keywords associated with clearly nonregulatory functions. This produced 250 candidate families, nArcR-1 through nArcR-250, comprising 13,237 proteins. Of the families, 171 have a characterized domain outside the reference set and 79 were uncharacterized by Pfam. Accordingly, the main novelty is the family-level organization and archaeal distribution of noncanonical clan members, not 250 newly discovered DNA-binding folds.

A previously completed 24-setting clustering sweep showed that 130 families remained stable across at least half the settings and 80 across at least 20 settings. Higher identity thresholds usually split families into pure subfamilies, whereas lower thresholds merged some boundaries. No family collapsed into a canonical cluster. Family existence is therefore more stable than exact granularity.

Within-panel resampling separates detection from abundance threshold sensitivity

We added a 500-replicate resampling analysis using the released panel and membership table. In stratified samples retaining 80 percent of each lineage, all 250 families were detected at least once in every replicate; a median of 212 retained membership in at least five sampled genomes (95 percent interval 204 to 220). When all rare groups were retained and the two capped groups were reduced to 80 percent, all families were again detected and a median of 224 retained five-genomes (218 to 231). A deliberately smaller panel capped every major group at 30 genomes, producing 157 genomes, recovered a median of 242 families at least once (235 to 247) and 137 at five-genomes (127 to 148). The analysis supports the robustness of catalog detection to moderate within-panel resampling, while showing that the five-genome inclusion rule is sensitive to panel size. It cannot substitute for full rediscovery from alternative draws of the unreleased 1,069-genome candidate pool. (Supplementary Fig. S1).

A separate lineage-stratified accumulation analysis sampled 10–100% of each major group. At 10% of the panel, the median sample contained 34 genomes and detected 187 families at least once, but only 54 in at least five genomes. At 50%, 249 families were detected and 156 met the five-genome criterion; all 250 were detected by the 60% samples, although only 174 met the five-genome criterion. The contrast shows that detecting a released family saturates earlier than satisfying the abundance threshold used during discovery. These values describe observation of fixed families and do not repeat discovery. (Supplementary Fig. S10).

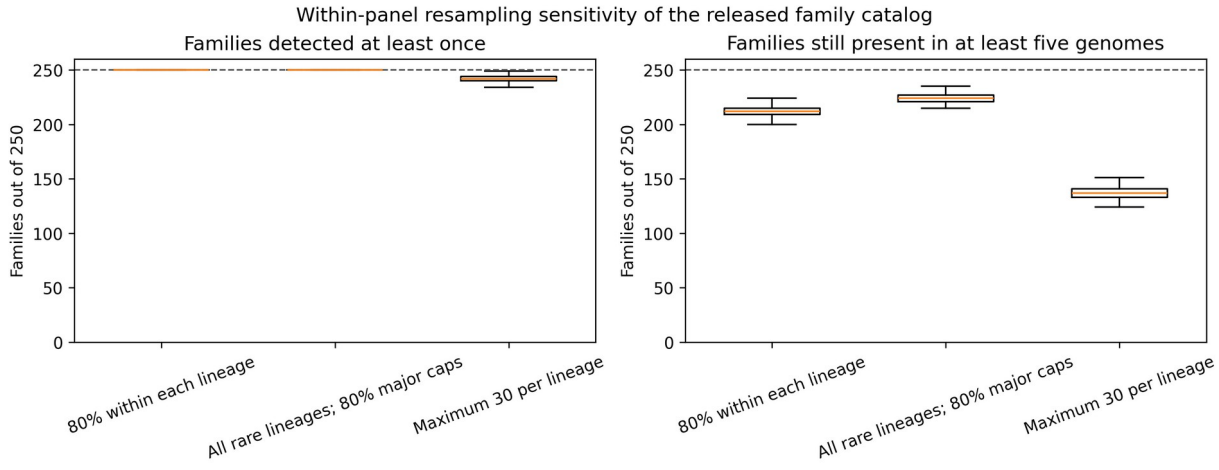


Figure 2. Within-panel resampling sensitivity. Boxes summarize 500 replicates per scheme. The dashed line marks the complete 250 family catalog. Family detection is robust to moderate stratified resampling, whereas retention above the five-genome criterion declines with sample size.

Family profiles are separable and distributions span broad to restricted

For 96 families with at least 20 members, 30% of sequences were withheld at random, family HMMs were rebuilt and the withheld sequences were reassigned. The correct family was recovered for 3,444 of 3,538 sequences (97.3%), with a median best-versus-second-best score margin of 85.9 bits. Because related sequences can occur in both partitions, this sequence-level split is an optimistic assignment check rather than phylogenetically independent validation.

We therefore added a catalog-wide complete-genus holdout analysis using all deposited sequences and the genus labels in the discovery-panel metadata. Normalized dipeptide centroids were fitted after removing every sequence from the test sequence's genus. The analysis evaluated 13,166 of 13,237 sequences from 245 of 250 families. Median family accuracy was 0.826, macro-averaged family accuracy was 0.721, and 197 families had a positive median own-family separation margin. Five families lacked sufficient genus-separated training data. This conservative composition-based test supports many, but not all, released boundaries and does not constitute calibrated HMM validation. (Supplementary Fig. S11).

Archived aggregate accounting places detections across 3,526 archaeal genomes: 1,324 isolates, 1,986 metagenome-assembled genomes and 216 single-amplified genomes. A rescue stage changed 379 genomes from an initial zero-hit status to at least one hit, and the released accounting flag has been corrected to match that final fate. Family-resolved presence-absence and copy-number matrices are available for only the earlier 3,192-genome pass. We therefore treat the 3,526-genome group summaries as archived descriptive aggregates and base accession-level prevalence, uncertainty and sensitivity analyses on the reproducible 3,192-genome matrix.

We tested profile transfer on accessions outside the discovery panel by drawing 500 lineage-matched samples of 331 genomes from the released 3,192 genome projection matrix. The samples used the discovery panel lineage counts and had zero accession overlap with the discovery set. A median of 245 of 250 families was detected at least once in each sample (95 percent interval 242 to 248), and a median of 206 occurred in at least five-genomes (198 to 215). This result supports transferability of the released family HMMs to independent genomes. It does not test whether the full scan and clustering workflow would rediscover the same family boundaries. (Supplementary Fig. S2).

We next stratified the accession-resolved 3,192-genome projection by genome origin, assembly level, and reported completeness/contamination. Each stratum retained 242–249 of the 250 families, except the smaller single-amplified-genome stratum, which retained 205. Median per-family prevalence varied from 0.025 in MAGs to 0.068 in isolates and from 0.034 in scaffold assemblies to 0.072 in complete/chromosome assemblies. These descriptive shifts are confounded by lineage and sampling, but they show why non-detection and cross-stratum prevalence differences must be interpreted conservatively. (Supplementary Fig. S4).

To reduce genome-level pseudoreplication, we inferred 533 genus labels from the deposited organism names and resampled genera as clusters for 1,000 replicates. Genome-weighted and genus-balanced prevalence differed appreciably for some families, and the versioned table reports both estimates with cluster-bootstrap intervals. Because the release lacks a versioned taxonomy and species tree, this is a taxonomic-cluster sensitivity analysis rather than formal phylogenetic comparative inference. (Supplementary Fig. S7).

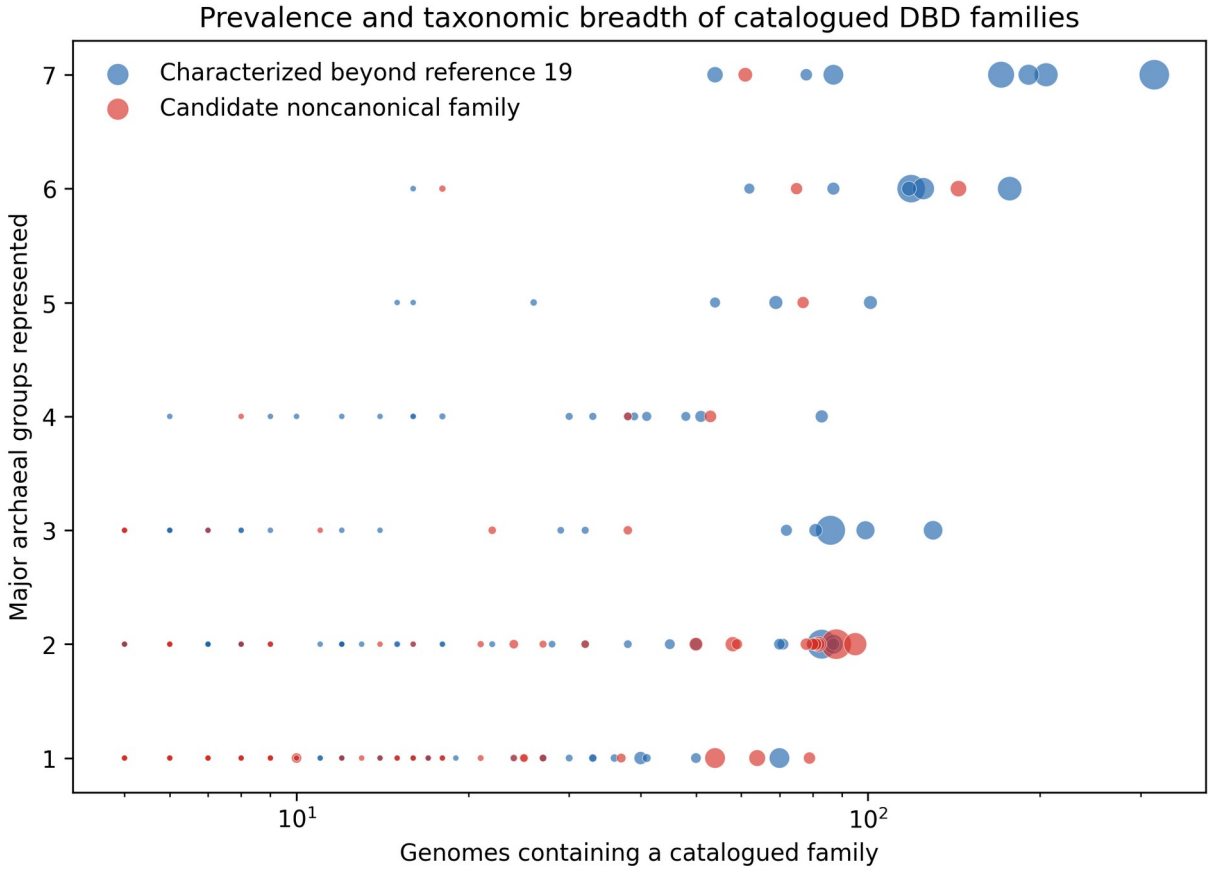


Figure 3. Prevalence and taxonomic breadth of catalogued DBD families. Point size reflects family membership. “Candidate noncanonical family” means outside the nineteen-family reference set within the two-clan search universe; it does not imply experimentally demonstrated regulatory function.

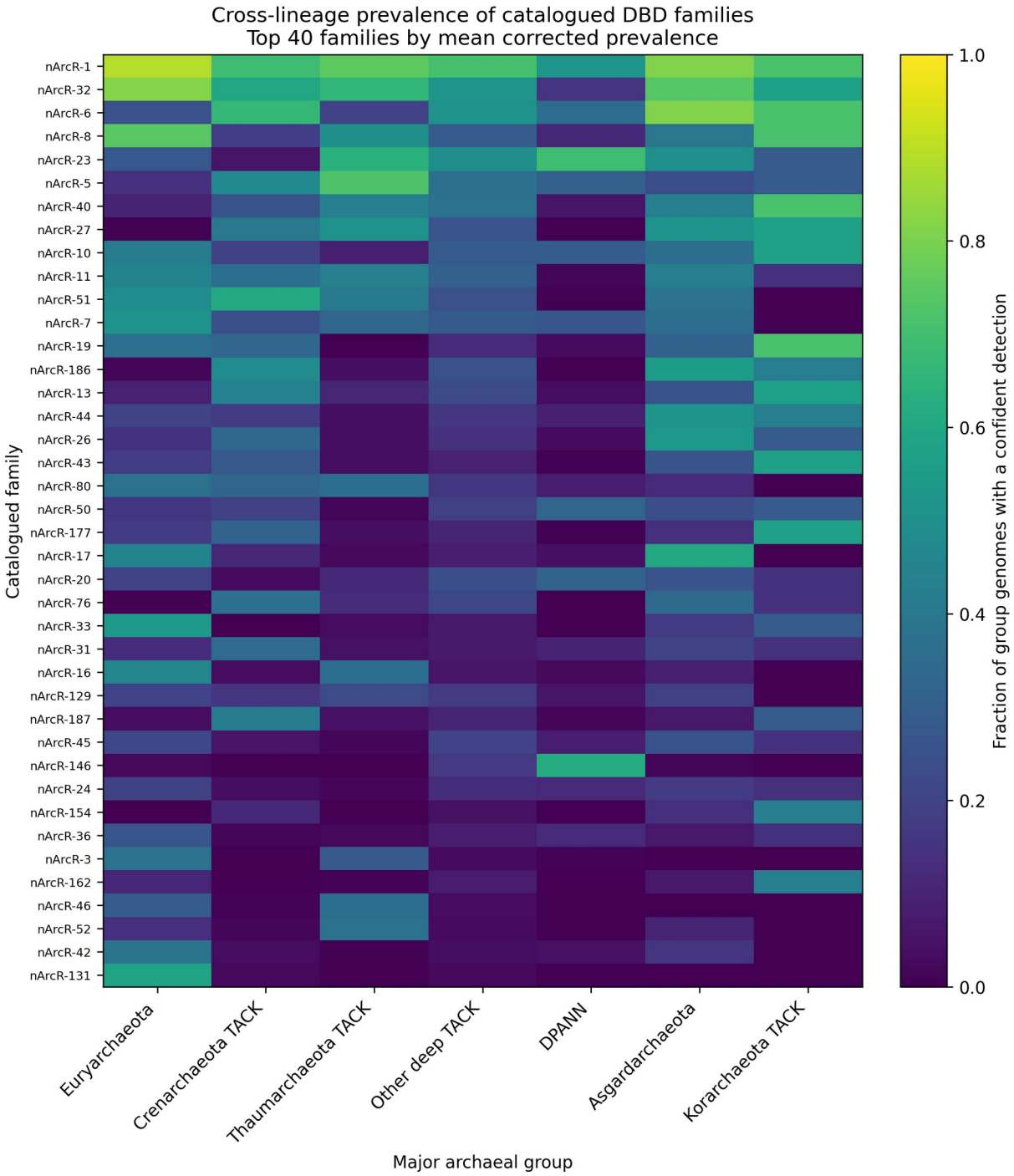


Figure 4. Archived cross-lineage prevalence for the 40 families with the highest mean prevalence. Values use aggregate 3,526-genome group denominators. Family-resolved matrices are unavailable for the final rescue stage, so these values are descriptive and are not used for accession-level uncertainty analyses.

Genomic neighborhoods generate hypotheses rather than confirmed regulons

An initial genome-shuffled enrichment screen identified at least one neighboring category for 178 families. The most frequent top categories included toxin-antitoxin systems, HalOD1-type signaling,

DNA repair and replication, transport, oxidoreductases, methyltransferases and metallohydrolases. The historical workflow also reported genus-collapsed and whole-neighborhood block permutations, but their accession-level output table was not preserved. We therefore treat the deposited initial enrichment table and Figure 5 as exploratory and do not present individual associations as dependence-corrected findings.

Upstream motif discovery provides a separate line of evidence. For each of the three null models, Benjamini-Hochberg adjustment was applied across the 12 tested families. Eleven families passed the mononucleotide, dinucleotide and shuffled-family-assignment nulls; nArcR-6 failed the assignment null. In the archived deeper analysis of six example families, five tests had nominal $p > 0.5$ and nArcR-20 had nominal $p = 0.04$. The source table needed to reconstruct correction across those six tests was not preserved, so the $p = 0.04$ result is not treated as significant. These results do not support a direct regulon claim.

We next relearned each motif after splitting genomes into 70 percent training and 30 percent test sets for 30 seeded replicates. All 12 motifs had median held-out AUC above 0.5 against dinucleotide-shuffled versions of the same sequences. Ten of 12 produced an AUC above 0.5 against GC-matched upstreams of other families in at least 26 resamples. nArcR-20 and nArcR-24 did not. The repeated splits overlap, so we report their distributions descriptively rather than treat them as independent observations for hypothesis testing. nArcR-6 showed weak but directionally consistent discrimination from other families (median AUC 0.580), which does not override its failure under the earlier shuffled family assignment null. These results provide out of sample sequence evidence for most tested motifs, but do not demonstrate protein binding or regulation. (Supplementary Fig. S3).

To reduce phylogenetic leakage further, we repeated motif learning while holding out complete genera. Train and test sets had zero genus overlap in every split. Seven of 12 families discriminated held-out family upstreams from GC-matched other-family upstreams in at least 26 of 30 resamples; eleven of 12 did so against dinucleotide-shuffled controls. The reduced support relative to genome-level splitting indicates that some apparent signal is lineage-associated and narrows the set of motifs prioritized for experiment. (Supplementary Fig. S5).

We then left out one major archaeal lineage at a time and used only same-lineage other-family controls. Ten of 12 families had at least one evaluable block under the minimum sample-size rules; nArcR-17 and nArcR-46 did not. Performance varied substantially among lineages, so block-level values rather than a pooled significance claim are reported. This is the most conservative deposited motif transfer analysis and further limits any general binding interpretation. (Supplementary Fig. S9).

Selected-family trees show taxonomic clustering

Maximum-likelihood trees were examined for ten prespecified example families containing 62–80 tips. Eight families had fewer observed phylum-state changes than expected in 500 tip-label permutations (permutation $p=0.002$ for each); nArcR-17 and nArcR-33 did not ($p=0.934$ and 0.904). Median ultrafast-bootstrap support

across the ten trees ranged from 86 to 98. These selected-family results show taxonomic-clustering but do not distinguish vertical inheritance from lineage-biased sampling or horizontal transfer.

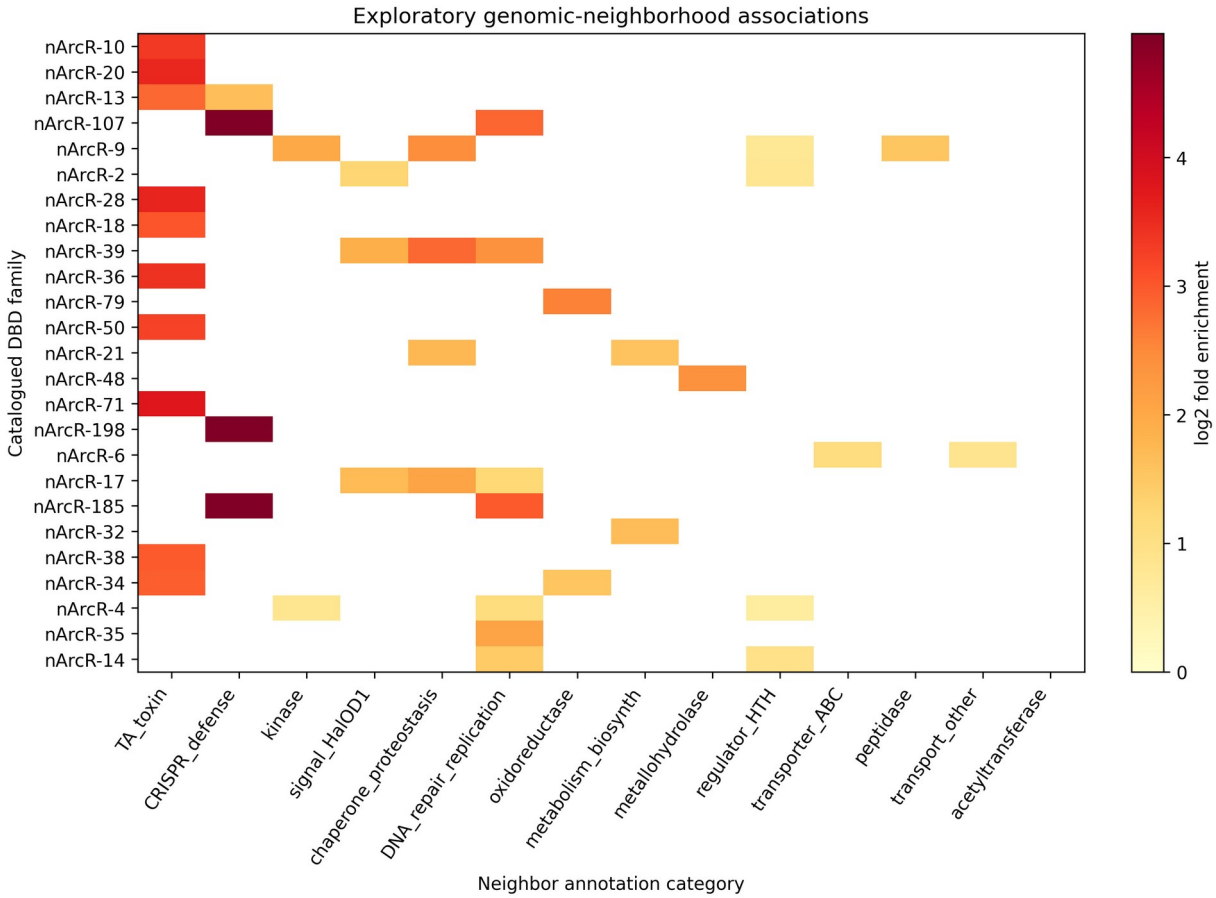


Figure 5. Exploratory genomic-neighborhood associations for selected catalogued families. Color encodes log2 fold enrichment in the initial deposited table. Associations generate functional hypotheses and do not establish direct transcriptional regulation; the historical dependence-corrected source table was not preserved.

Most apparently dark families resolve to known sequence or fold classes

AlphaFold Database models were located for 193 representatives, and 184 produced scored Foldseek comparisons against 66 curated DNA-binding and non-DNA-binding reference folds. At the prespecified TM-score threshold of 0.55, 175 of 184 scored representatives matched a DNA-binding reference and nine did not; a different nine families lacked a scored comparison. Because the catalog was discovered inside HTH/RHH Pfam clans and the reference library was curated rather than exhaustive, this is a structural-consistency analysis rather than independent proof of DNA binding or fold novelty.

Among the 79 families initially uncharacterized by Pfam, Conserved Domain Database searches classified 56. Fourteen additional families without a CDD classification exceeded the structural threshold, and nine had neither a qualifying CDD result nor a qualifying structural match. The sequence and structural counts refer to this 79-family decision flow; they are distinct from the nine catalog representatives without a scored Foldseek comparison.

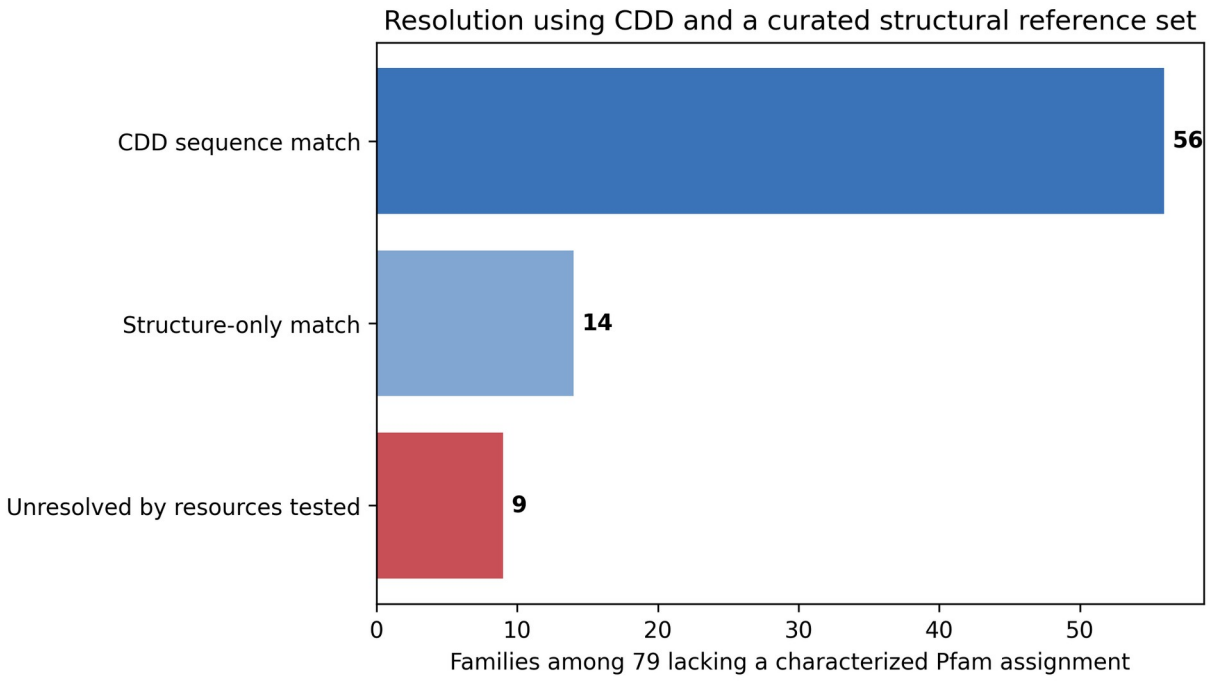


Figure 6. Reassessment of the 79 Pfam uncharacterized families using the Conserved Domain Database and structural comparisons. Most are resolved by broader sequence databases or a known fold; nine remain unresolved by the resources tested.

Discussion

The two targeted Pfam clans contain substantial archaeal diversity outside the historical nineteen-family reference assignment. That excess is not a collection of isolated sequences: it forms separable families, many recur across genomes and lineages, and most representatives map to recognized domain or fold classes. The strongest conclusion is therefore that familiar annotations underdescribe the family structure of archaeal HTH and RHH clan proteins.

The result is narrower than the original framing of a domain-wide map of transcription factors. The workflow begins with CL0123 and CL0057, so it cannot discover DNA-binding proteins outside those clans or proteins too diverged to match their profiles. It also cannot infer transcription factor activity from a domain alone. Recasting the study around a bounded catalog makes its quantitative claims testable and its novelty more precise.

The 331-genome panel is the deterministic output of a lineage-balancing rule, except for the seeded proportional samples within the two capped groups. Its purpose is to prevent heavily sequenced clades from dominating family discovery while retaining all genomes from five underrepresented groups. The particular caps remain pragmatic. Within-panel resampling shows that moderate reductions preserve detection of every family, but smaller panels disproportionately remove families near the five-genome threshold. Independent projection samples recovered a median of 245 families, showing that most profiles transfer beyond the

discovery accessions. A stronger future test should rerun the complete scan and clustering workflow on alternative panels drawn from the full candidate pool and quantify family matching across runs.

The evidence is strongest for the existence of recurring sequence groups, although the complete-genus analysis shows that family separability is heterogeneous and weaker than random sequence holdout suggests. Structural comparisons are consistent with known DNA-binding folds for most scored representatives but are not independent of the two-clan discovery design. Genomic context suggests biological settings, while genus- and lineage-blocked motif results limit direct regulatory interpretation. In the selected phylogenetic subset, eight of ten families showed taxonomic-clustering; that pattern is compatible with several evolutionary processes and does not identify one.

The nine families unresolved by both CDD and the curated structural library provide the clearest experimental priorities. Exploratory neighborhood associations can guide locus selection but are not used as dependence-corrected evidence. Electrophoretic mobility shift assays, DNA-affinity measurements, knockouts or CRISPR interference, and transcriptome profiling after perturbation would establish which families are transcription factors, their operators and their direct targets.

First, the search universe is restricted to two Pfam clans, and the exact historical Pfam release and nineteen-family/twenty-model reference inventory were not preserved. Second, the 5-bit reciprocal-exclusion tolerance, lineage caps and nonregulatory keyword filter cannot be evaluated by full de novo sensitivity analysis because the raw profile hits and 1,069-genome candidate pool are unavailable. Third, the final 3,526-genome family-resolved matrices are missing; only aggregate summaries and the accession-resolved 3,192-genome matrices are reproducible. Fourth, dependence-corrected neighborhood and deep-motif source tables are absent, so those results remain exploratory. Fifth, computational motifs and structural matches do not replace binding assays. Sixth, prevalence strata differ in taxonomic composition. Seventh, genus and major-lineage motif holdouts attenuate several signals. Eighth, genus labels were inferred from organism names rather than a preserved taxonomy. Ninth, the new catalog-wide complete-genus analysis is composition based; a fully calibrated blocked-HMM benchmark remains future work.

A search bounded to two archaeal HTH and RHH Pfam clans produced 250 released sequence families outside an historical reference assignment. Complete-genus holdout supports many family boundaries but also reveals heterogeneous separability. Most initially uncharacterized families map to broader sequence or structural classes, while nine remain unresolved by the resources tested. The catalog is a versioned hypothesis set for experiments, not a comprehensive archaeal regulome or a set of confirmed transcription factors.

Methods

Study design and interpretation

This computational comparative-genomics study has a primary descriptive objective: document a bounded historical family catalog within Pfam clans CL0123 and CL0057.

Random sequence holdout, independent-accession projection and the catalog-wide complete-genus composition analysis are secondary validation analyses. Panel resampling, catalog accumulation, genome-quality stratification, clustered prevalence and taxonomically blocked motif tests are robustness analyses. Neighborhood, phylogenetic and structural interpretations are exploratory. These labels describe the evidence hierarchy; the analyses were not preregistered before the underlying catalog was examined.

Genome panel assembly

Archaeal assemblies were assigned to seven mutually exclusive historical major-group categories: Euryarchaeota, Crenarchaeota within TACK, Thaumarchaeota within TACK, other deep TACK, Korarchaeota within TACK, DPANN and Asgardarchaeota. All genomes in five underrepresented categories were retained. Euryarchaeota and Crenarchaeota within TACK were capped at 140 and 70; subgroup allocations were proportional and genomes were sampled without replacement with seed 42. The exact GTDB release was not preserved, so these labels describe the deposited grouping rather than a current taxonomic reclassification.

Targeted domain scan and reciprocal exclusion

Pfam profiles in clans CL0123 and CL0057 defined the historical search universe. The source analysis reports twenty models representing nineteen reference families and 747 candidate profiles, but their exact accessions, family mapping and Pfam release were not preserved. HMMER searches used each available model's gathering threshold. For every protein, the highest-domain-score candidate interval was compared with the highest-scoring reference hit. Interval overlap was shared aligned residues divided by candidate-interval length. A candidate was excluded when overlap exceeded 0.5 and the reference score was at least the candidate score minus 5 bits. The highest-scoring retained candidate was used when multiple candidates remained. Supplementary Table S3 records which search-universe identifiers are preserved and which are unrecoverable.

Family construction and sensitivity analyses

Retained proteins were clustered with MMseqs2 using minimum sequence identity 0.30, coverage 0.60, coverage mode 0 and cluster mode 1 [19,20]. Clusters required a median length below 250 amino acids, at least 60% of members shorter than 250 amino acids and occurrence in at least five-genomes. Clusters whose dominant Pfam label matched the explicit nonregulatory exclusion list in Supplementary Methods were removed. The clustering sensitivity analysis varied identity from 0.25 to 0.40, coverage from 0.50 to 0.80 and clustering mode across 24-settings.

For the new panel sensitivity analysis, genomes were resampled without replacement for 500 replicates under three schemes: 80 percent of every lineage; all rare lineages plus 112 Euryarchaeota and 56 Crenarchaeota; and a maximum of 30 genomes per lineage. For every replicate we counted catalog families detected at least once and families retaining at least five-genomes. This analysis uses fixed released family memberships and therefore measures catalog retention, not de novo rediscovery. (Supplementary Fig. S1).

Family projection and validation

A profile HMM was built for each family. For 96 families with at least 20 members, 30% of sequences were removed at random, HMMs were rebuilt and withheld sequences were assigned by best score. This split was not taxonomically blocked. Projection used the fixed family HMM library. Accession-resolved analyses use the 3,192-genome matrix; the final 3,526-genome values are retained only as archived aggregate accounting.

For the catalog-wide blocked analysis, every deposited protein was represented by normalized dipeptide frequencies. For each test genus, all sequences from that genus were removed before family centroids were fitted. A sequence was evaluated only when its family retained at least two training sequences. Cosine similarity assigned the best of all evaluable family centroids. We report sequence-level outcomes, exact family binomial intervals, macro- and micro-accuracy, separation margins and unevaluable families. This alignment-free test does not calibrate HMM thresholds.

For independent genome validation, discovery panel accessions were removed from the legacy 3,192-genome presence-absence matrix. Five hundred samples were then drawn without replacement using the discovery panel lineage counts: 140 Euryarchaeota, 70 Crenarchaeota (TACK), 51 Thaumarchaeota, 33 DPANN, 19 Asgardarchaeota, 15 other deep TACK, and 3 Korarchaeota. For each sample we counted families detected at least once and in at least five-genomes. The procedure used fixed family HMM projections and therefore measures transferability rather than de novo discovery. (Supplementary Fig. S2).

For the quality sensitivity analysis, projection genomes were grouped by isolate, metagenome-assembled genome, or single-amplified genome status; by complete/chromosome, scaffold, or contig assembly; and by whether reported completeness was at least 90 percent with contamination at most 5 percent. We reported denominators, per-family prevalence, and the number of detected families for each stratum. No inferential comparison was made because these variables are confounded with lineage and database sampling. (Supplementary Fig. S4).

For clustered prevalence uncertainty, genus labels were inferred from deposited organism names, with Candidatus names retaining the following token. Genera were sampled with replacement for 1,000 seeded replicates. We calculated genus-balanced prevalence and a cluster-weighted prevalence interval for each family. This analysis addresses clustered sampling but does not model a species phylogeny.

Genomic neighborhood and motif analyses

Genes within five coding sequences of family members were assigned functional categories. Initial family-category enrichment used genome-conditioned label sampling, binomial tests and Benjamini-Hochberg correction [35]. The complete tested family-category universe was not retained, so the correction cannot be independently reconstructed. The historical workflow also described genus collapse and 2,000 whole-neighborhood block permutations with seed 42, but their source table was not preserved. No individual neighborhood association is therefore treated as a reproducible dependence-corrected finding. Up to 250

strand-aware upstream nucleotides were used for motif discovery under a zero-or-one occurrence model. For each null model, false-discovery-rate correction was applied across the 12 families. Six example-family motifs were also assessed with 1,000 permutations, but the source table required to reconstruct correction across those six tests is unavailable.

For the genome-level motif holdout analysis, genomes within each of 12 families were randomly divided into 70 percent training and 30 percent testing sets for 30 seeded replicates. A position weight matrix was learned only from training sequences under a zero or one occurrence model and scored in both orientations on held-out upstreams. Each held-out set was compared with dinucleotide-shuffled versions of the same sequences and with equally sized, GC-matched upstream sets from other families. Discrimination was summarized by area under the receiver operating characteristic curve, its 2.5 and 97.5 percent resampling quantiles, and the fraction of splits with AUC above 0.5. Because the splits overlap, no inferential test treats them as independent observations.

For genus-blocked validation, genera rather than genomes were randomly assigned to the 70/30 training and testing partitions. All upstream sequences from a held-out genus remained in the test set. The same PWM fitting, dinucleotide-shuffle control, GC-matched other-family control, AUC summaries, 30 replicate count, and descriptive interpretation were used. Train/test genus overlap was checked and was zero for every family and replicate. (Supplementary Fig. S5).

For leave-one-major-lineage-out validation, each lineage with at least five target sequences was held-out in turn. Training required at least twenty sequences, and the comparison required enough other-family upstreams from the same held-out lineage. A PWM learned from all remaining lineages was evaluated against dinucleotide-shuffled targets and GC-matched same-lineage controls. Blocks failing any minimum were reported as unevaluable rather than pooled. (Supplementary Fig. S9).

Phylogenetic and structural analyses

Ten previously selected example families with 62–80 deposited tips were aligned with MAFFT, trimmed with trimAl and analyzed with IQ-TREE 2 using ModelFinder and 1,000 ultrafast-bootstrap replicates [22–27]. The release does not preserve a prospective rule for selecting these ten families, so the analysis is treated as exploratory. Phylum clustering was measured by Fitch parsimony and compared with 500 tip-label shuffles. AlphaFold Database structures were located for 193 representatives; 184 yielded scored Foldseek comparisons against 66 curated DNA-binding and non-DNA-binding references [30–32]. A TM-score of at least 0.55 was used as a descriptive match threshold. Representatives of the 79 Pfam-uncharacterized families were also searched against CDD version 3.21 at E-value 0.01.

Ethics statement

This computational study used publicly available genomic and structural data and involved no human participants, identifiable human data, human tissue or live animals. Institutional ethics approval and informed consent were therefore not applicable.

Reproducibility, software and use of generative AI

The deposited reproduction environment uses Python 3.12 with NumPy 2.0.2, pandas 2.3.3, SciPy 1.13.1 and Matplotlib 3.9.4, and pins HMMER 3.4, MMseqs2 15.6f452, MAFFT 7.526, trimAl 1.4.1, IQ-TREE 2.3.6 and Foldseek 9.427df8a. Exact historical Pfam and GTDB releases, database download dates and the complete genome-download manifest were not preserved in the supplied source archive; this provenance gap is documented and no version was inferred retrospectively. Generative artificial-intelligence tools assisted language editing, code drafting, reproducibility packaging and document formatting. The authors inspected the underlying data, executed and checked the analyses, verified citations and outputs, and take responsibility for the final text, code, results and conclusions.

Data availability

The deposited datasets and versioned analysis outputs are available from Zenodo under the stable concept DOI https://doi.org/10.5281/zenodo.23461089 [41]. Version 0.5.0 is archived at https://doi.org/10.5281/zenodo.23496681. The release includes all 13,237 catalog-member sequences, family membership, discovery-panel metadata, final genome accounting, the 3,192-genome family matrices, validation outputs and explicit provenance tables. The exact historical profile inventory, raw discovery hits, complete candidate pool, corrected 3,526-genome family matrices and several dependence-corrected source tables remain unavailable. The 3,526-genome values are therefore identified as archived aggregate results.

Code availability

Custom code, automated tests, locked software environments, a container recipe, checksums and worked examples are available under the MIT License at https://github.com/ajayvellanki0720-boop/archaeal-dbd-catalog. The immutable version 0.5.0 archive is available at https://doi.org/10.5281/zenodo.23496681 [41]; the GitHub release records later corrections and catalog-wide blocked validation.

References

1. Aravind, L. DNA-binding proteins and evolution of transcription regulation in the archaea. *Nucleic Acids Research* **27**, 4658–4670 (1999). https://doi.org/10.1093/nar/27.23.4658

2. Dodd, I. B. & Egan, J. B. Improved detection of helix-turn-helix DNA-binding motifs in protein sequences. *Nucl Acids Res* **18**, 5019–5026 (1990). https://doi.org/10.1093/nar/18.17.5019

3. Schreiter, E. R. & Drennan, C. L. Ribbon–helix–helix transcription factors: variations on a theme. *Nat Rev Microbiol* **5**, 710–720 (2007). https://doi.org/10.1038/nrmicro1717

4. Guerra, A. J., Dann III, C. E. & Giedroc, D. P. Crystal Structure of the Zinc-Dependent MarR Family Transcriptional Regulator AdcR in the Zn(II)-Bound State. *J. Am. Chem. Soc.* **133**, 19614–19617 (2011). https://doi.org/10.1021/ja2080532

5. Thaw, P. Structural insight into gene transcriptional regulation and effector binding by the Lrp/AsnC family. *Nucleic Acids Research* **34**, 1439–1449 (2006). https://doi.org/10.1093/nar/gkl009

6. Reichelt, R., Gindner, A., Thomm, M. & Hausner, W. Genome-wide binding analysis of the transcriptional regulator TrmBL1 in Pyrococcus furiosus. *BMC Genomics* **17**, 40 (2016). https://doi.org/10.1186/s12864-015-2360-0

7. Makarova, K. S., Wolf, Y. I. & Koonin, E. V. Comprehensive comparative-genomic analysis of Type 2 toxin-antitoxin systems and related mobile stress response systems in prokaryotes. *Biol Direct* **4**, 19 (2009). https://doi.org/10.1186/1745-6150-4-19

8. Makarova, K. S., Grishin, N. V. & Koonin, E. V. The HicAB cassette, a putative novel, RNA-targeting toxin-antitoxin system in archaea and bacteria. *Bioinformatics* **22**, 2581–2584 (2006). https://doi.org/10.1093/bioinformatics/btl418

9. Arcus, V. L., McKenzie, J. L., Robson, J. & Cook, G. M. The PIN-domain ribonucleases and the prokaryotic VapBC toxin-antitoxin array. *Protein Engineering Design and Selection* **24**, 33–40 (2011). https://doi.org/10.1093/protein/gzq081

10. Liu, T. et al. Coupling transcriptional activation of CRISPR–Cas system and DNA repair genes by Csa3a in Sulfolobus islandicus. *Nucleic Acids Research* **45**, 8978–8992 (2017). https://doi.org/10.1093/nar/gkx612

11. Sorek, R., Lawrence, C. M. & Wiedenheft, B. CRISPR-Mediated Adaptive Immune Systems in Bacteria and Archaea. *Annu. Rev. Biochem.* **82**, 237–266 (2013). https://doi.org/10.1146/annurev-biochem-072911-172315

12. Xie, R. et al. Expanding Asgard members in the domain of Archaea sheds new light on the origin of eukaryotes. *Sci. China Life Sci.* **65**, 818–829 (2022). https://doi.org/10.1007/s11427-021-1969-6

13. Dombrowski, N. et al. Undinarchaeota illuminate DPANN phylogeny and the impact of gene transfer on archaeal evolution. *Nat Commun* **11**, 3939 (2020). https://doi.org/10.1038/s41467-020-17408-w

14. Allers, T., Ngo, H. P., Mevarech, M. & Lloyd, R. G. Development of Additional Selectable Markers for the Halophilic Archaeon Haloferax volcanii Based on the leuB and trpA Genes. *Appl Environ Microbiol* **70**, 943–953 (2004). https://doi.org/10.1128/AEM.70.2.943-953.2004

15. Chaumeil, P. A., Mussig, A. J., Hugenholtz, P. & Parks, D. H. GTDB-Tk: a toolkit to classify genomes with the Genome Taxonomy Database. *Bioinformatics* **36**, 1925–1927 (2020). https://doi.org/10.1093/bioinformatics/btz848

16. O'Leary, N. A. et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Res* **44**, D733–D745 (2016). https://doi.org/10.1093/nar/gkv1189

17. Eddy, S. R. Accelerated Profile HMM Searches. *PLoS Comput Biol* **7**, e1002195 (2011). https://doi.org/10.1371/journal.pcbi.1002195

18. Mistry, J. et al. Pfam: The protein families database in 2021. *Nucleic Acids Research* **49**, D412–D419 (2021). https://doi.org/10.1093/nar/gkaa913

19. Steinegger, M. & Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat Biotechnol* **35**, 1026–1028 (2017). https://doi.org/10.1038/nbt.3988

20. Steinegger, M. & Söding, J. Clustering huge protein sequence sets in linear time. *Nat Commun* **9**, 2542 (2018). https://doi.org/10.1038/s41467-018-04964-5

21. Buchfink, B., Reuter, K. & Drost, H. G. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat Methods* **18**, 366–368 (2021). https://doi.org/10.1038/s41592-021-01101-x

22. Katoh, K. & Standley, D. M. MAFFT Multiple Sequence Alignment Software Version 7: Improvements in Performance and Usability. *Molecular Biology and Evolution* **30**, 772–780 (2013). https://doi.org/10.1093/molbev/mst010

23. Capella-Gutiérrez, S., Silla-Martínez, J. M. & Gabaldón, T. trimAl: a tool for automated alignment trimming in large-scale phylogenetic analyses. *Bioinformatics* **25**, 1972–1973 (2009). https://doi.org/10.1093/bioinformatics/btp348

24. Price, M. N., Dehal, P. S. & Arkin, A. P. FastTree 2 – Approximately Maximum-Likelihood Trees for Large Alignments. *PLoS ONE* **5**, e9490 (2010). https://doi.org/10.1371/journal.pone.0009490

25. Minh, B. Q. et al. IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era. *Molecular Biology and Evolution* **37**, 1530–1534 (2020). https://doi.org/10.1093/molbev/msaa015

26. Kalyaanamoorthy, S., Minh, B. Q., Wong, T. K. F., von Haeseler, A. & Jermiin, L. S. ModelFinder: fast model selection for accurate phylogenetic estimates. *Nat Methods* **14**, 587–589 (2017). https://doi.org/10.1038/nmeth.4285

27. Hoang, D. T., Chernomor, O., von Haeseler, A., Minh, B. Q. & Vinh, L. S. UFBoot2: Improving the Ultrafast Bootstrap Approximation. *Molecular Biology and Evolution* **35**, 518–522 (2018). https://doi.org/10.1093/molbev/msx281

28. Bailey, T. L., Johnson, J., Grant, C. E. & Noble, W. S. The MEME Suite. *Nucleic Acids Res* **43**, W39–W49 (2015). https://doi.org/10.1093/nar/gkv416

29. Hyatt, D. et al. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinformatics* **11**, 119 (2010). https://doi.org/10.1186/1471-2105-11-119

30. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021). https://doi.org/10.1038/s41586-021-03819-2

31. Varadi, M. et al. AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research* **50**, D439–D444 (2022). https://doi.org/10.1093/nar/gkab1061

32. van Kempen, M. et al. Fast and accurate protein structure search with Foldseek. *Nat Biotechnol* **42**, 243–246 (2024). https://doi.org/10.1038/s41587-023-01773-0

33. Kunzmann, P. & Hamacher, K. Biotite: a unifying open source computational biology framework in Python. *BMC Bioinformatics* **19**, 346 (2018). https://doi.org/10.1186/s12859-018-2367-z

34. Liao, Y., Smyth, G. K. & Shi, W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* **30**, 923–930 (2014). https://doi.org/10.1093/bioinformatics/btt656

35. Benjamini, Y. & Hochberg, Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **57**, 289–300 (1995). https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

36. Harris, C. R. et al. Array programming with NumPy. *Nature* **585**, 357–362 (2020). https://doi.org/10.1038/s41586-020-2649-2

37. Virtanen, P. et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat Methods* **17**, 261–272 (2020). https://doi.org/10.1038/s41592-019-0686-2

38. McKinney, W. Data Structures for Statistical Computing in Python. *Proceedings of the Python in Science Conference*, 56–61 (2010). https://doi.org/10.25080/Majora-92bf1922-00a

39. Seabold, S. & Perktold, J. Statsmodels: Econometric and Statistical Modeling with Python. *Proceedings of the Python in Science Conference*, 92–96 (2010). https://doi.org/10.25080/Majora-92bf1922-011

40. Cock, P. J. A. et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics* **25**, 1422–1423 (2009). https://doi.org/10.1093/bioinformatics/btp163

41. Vellanki, A. & Dosapati, S. Archaeal DBD catalog: reproducibility package (version 0.5.0). Zenodo https://doi.org/10.5281/zenodo.23496681 (2026).

Acknowledgements

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author contributions

AV and SD jointly contributed to conceptualization, methodology, investigation, formal analysis, validation, visualization, data curation, software and writing. Both authors reviewed and approved the final manuscript.

Additional information

Competing interests

The authors declare no competing interests.